

Assignment #1

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CMS# S24-BSSE-0004

Artificial Intelligence

Question #1

- If a student Passes the midterm and final exam, the student Pass the course.
- If a student Passes the course, the student is eligible for a certificate
- Ali Passes the midterm exam and final exam.

a. Representation facts and rules using Propositional logic.

A : Passes the midterm

B : Passes the Final exam

C : Passes the course

D : Eligible for a certificate

1. Student Passes Mids and Final, the Student Passes the course

$$(A \wedge B) \rightarrow C$$

2. Student Passes the course,

Student is eligible for certificate

$$C \rightarrow D$$

3. Ali Passes the mid and Finals

$$A \wedge B$$

b. Inference To determine whether Ali is eligible

Step 1: $A \wedge B$

Step 2: $(A \wedge B) \rightarrow C$

Step 3: $C \rightarrow$ modus Ponens of steps 1 or 2

Step 4: $C \rightarrow D$

Step 5: $D \rightarrow$ modus Ponens of steps 3, 4

So, this inference where Ali is eligible for certificate.

Question #2

$$(P \rightarrow Q) \wedge (Q \rightarrow R)$$

a. Convert the expression into an equivalent

$$P \rightarrow Q = \sim P \vee Q$$

$$Q \rightarrow R = \sim Q \vee R$$

$$(P \rightarrow Q) \wedge (Q \rightarrow R) = (\sim P \vee Q) \wedge (\sim Q \vee R)$$

b. logical Equivalence mean.

logical Equivalence means that two logical expression have same truth table in every possible situation

$$P \rightarrow Q = \sim Q \rightarrow \sim P$$

c. $(P \rightarrow Q) (\sim Q \rightarrow \sim P)$ justify.

$$(P \rightarrow Q) = (\sim P \vee Q)$$

P	Q	$\sim P \vee Q$	$\sim Q \rightarrow \sim P$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

Question #3

$P \vee Q \wedge \sim P$ through Table

P	Q	$P \vee Q$	$\sim P$	$P \vee Q \wedge \sim P$
T	T	T	F	F
T	F	T	F	F
F	T	T	T	T
F	F	F	T	F

So, the result show that, it is neither a tautology nor a contradiction.

Question #4

Robotic vacuum cleaner
operating in a house.

a. Identify

Sensor: Dirt sensor, drop sensor,
bump sensor, wheel encoders
and camera or Lidar.

Actuators: wheels / motors, brush,
suction fan, charging connector

Percepts: Dirt detected or not,
Obstacle ahead, edge of stairs.

Actions: Move forward/back,
Turn left/right, start or stop
suction, Return to dock stop.

b. PEAS

Performance: amount of dirt cleaned
area covered, battery efficiency
Time Taken, no collision or fails.

Environment: Rooms, floor, carpets
furniture, stairs.

Actuators: wheels, brush, vacuum
dock connector.

Sensors: Dirt, bump, cliff, camera, lidar, battery sensor.

C.

Partially observable: It only senses what's near it and can't see dirt in other rooms.

Stochastic: Dirt appears randomly and people or pets move around, so outcomes aren't fully predictable.

Sequential: Current moves (like battery use and cleaned areas) affect future decisions.

Dynamic: The environment changes while the robot is working.

Continuous: Positions, speeds, and battery level take continuous values.

Question #5

- P: It is raining
- Q: The road is wet
- R: The school is closed.

a. Statements into Proposition logic.

$$P \rightarrow Q$$

$$R \leftrightarrow P$$

$$\sim Q \rightarrow \sim P$$

b. Construct a Complete Truth Table

$$(P \rightarrow Q) \wedge (Q \rightarrow R)$$

P	Q	R	$\sim P$	$\sim Q$	$\sim P \vee Q$	$\sim Q \vee R$	$(P \rightarrow Q) \wedge (Q \rightarrow R)$
T	T	T	F	F	T	T	T
T	T	F	F	F	T	F	F
T	F	T	F	T	F	T	F
T	F	F	F	T	F	T	F
F	T	T	T	F	T	T	T
F	T	F	T	F	T	F	F
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T